

Was magnitude estimation necessary?

A secondary-data reassessment of Bard, Robertson, and Sorace (1996)

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Abstract

This paper reassesses Bard, Robertson, and Sorace’s case for magnitude estimation in linguistic acceptability research. It separates two claims that are often run together: acceptability judgments are graded, measurable, and methodologically serious; magnitude estimation has a practical advantage over simpler formal judgment tasks. The first claim is the stronger legacy. The second requires later method-comparison evidence. The paper treats Sprouse comparison datasets as the row-level basis for constrained secondary analysis, uses Langsford et al. as article-level evidence about reliability and response-style risks, and keeps computational benchmarks separate as afterlife evidence rather than evidence about magnitude estimation itself.

Keywords: magnitude estimation; acceptability judgments; grammaticality judgments; secondary data; experimental syntax

I THE PROBLEM

Bard, Robertson, and Sorace made a methodological claim that was larger than a single elicitation format. Acceptability judgments, they argued, could be treated as graded measurements rather than as informal binary verdicts (Bard et al., 1996). That move helped make experimental syntax a measurement enterprise: judgments could be collected from naive participants, scaled, compared across groups, and subjected to ordinary statistical tests.

The harder question is whether magnitude estimation was necessary for that advance. Magnitude estimation promised a way around the scale restrictions of traditional judgment tasks, giving participants an open response scale and preserving fine distinctions. But later acceptability work did not settle into one privileged elicitation method. Likert ratings, forced choice, yes/no tasks, and Thurstonian models all became part of the measurement toolkit (Langsford et al., 2018; Sprouse & Almeida, 2017; Sprouse et al., 2013).

This paper separates two claims that are often allowed to travel together. One claim is strong and still plausible: acceptability is graded, measurable, and methodologically serious. The other is

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weaker: magnitude estimation has a practical measurement advantage over simpler formal tasks. The first claim can survive even if the second does not.

The reassessment is therefore not a rejection of Bard et al.'s programme. It is a test of what should be credited to magnitude estimation itself, and what should instead be credited to the broader formalization of acceptability judgment methodology.

2 THE BARD TARGET

The target has to be reconstructed before it is reassessed. Bard et al. present magnitude estimation as a solution to measurement-scale problems in conventional acceptability elicitation, with enough resolution to capture fine-grained linguistic distinctions and enough robustness to work with linguistically naive participants (Bard et al., 1996). That is the source-grounded claim this paper can evaluate against later method-comparison evidence.

A stronger claim sometimes follows in reception: magnitude estimation is not just one formal method among others, but the method with privileged measurement status. That stronger reading should not be attributed to Bard et al. without textual support. It is better treated as a separable claim about what parts of the field took magnitude estimation to have shown.

The distinction matters for fairness and for design. If Bard et al. are read as saying only that acceptability judgments need a graded measurement method, then later Likert, forced-choice, yes/no, and Thurstonian work may extend their programme rather than refute it. If magnitude estimation is read as having a method-specific privilege, then later comparisons have to ask whether it outperforms other tasks on prediction, reliability, decision stability, or construct interpretation.

The available evidence also limits the claim. No public Bard participant-level rows have been found, so this paper cannot directly replicate the original experiment. The empirical reassessment has to be a conceptual replication and secondary-data test: later datasets are used to ask whether the method-specific part of the Bard claim retains independent evidential force.

3 SECONDARY-DATA DESIGN

The secondary-data design treats the project as a measurement reassessment of Bard, Robertson, and Sorace (Bard et al., 1996). Four evidential routes should stay separate. Published summaries from Bard et al. can source the historical claim, but not a direct replication unless their participant-level data are found.

Later judgment studies can support secondary analysis when they compare elicitation methods on shared or comparable items and provide the rows needed for the intended estimand. Computational benchmarks can show the afterlife of acceptability labels, but they do not bear on magnitude estimation versus Likert ratings. Construct-level discussion can separate acceptability from grammaticality, naturalness, standardness, and correction.

This creates a genre gate. Public row-level data from the Sprouse comparison studies can support a constrained secondary analysis of method performance. The Langsford et al. reliability study is essential evidence for the argument, but unless its raw participant responses are obtained it can only constrain the paper at the article-summary level. The paper should therefore not promise a full response-style or test-retest model across the whole literature.

The Sprouse files also require representation choices before modelling. For Sprouse et al. (2013), the primary forced-choice comparison should use the pair-level sign-test file, because forced choice

Source	Evidence level	Use in this paper	Claim it cannot support
Bard et al. 1996	Published article	Historical target and scale-resolution claim	Direct replication without participant rows
Sprouse 2013/2017	Public row-level comparison data	Constrained secondary method comparison, pending reuse permission	Ungated publication-ready novel results while reuse is pending
Langsford et al. 2018	Published article and supplement	Reliability and response-style synthesis	New participant-level reliability model without raw rows
CoLA, BLiMP, SAD	Benchmark labels and documentation	Afterlife of formal acceptability measurement	Evidence about ME versus other human judgment methods

Table 1: Evidential routing before analysis. ME = magnitude estimation; SAD = Syntactic Acceptability Dataset.

is a pairwise task and that file already records judgments against expected patterns. The logistic file preserves item-level binary rows, changing the unit of analysis, so it belongs as a named sensitivity specification.

For Sprouse and Almeida (2017), yes/no rows with item IDs B, G, and M should be excluded from item-level comparisons. They are absent from the materials workbook, occur in repeated early order positions for every subject, and do not align with the magnitude-estimation, Likert, or forced-choice item universe. This exclusion is a structural inclusion rule, not an outcome-based trimming decision.

Before any modelling, each source should be assigned to an estimand it can actually answer. The primary estimand is the item-level acceptability signal recoverable across methods. Secondary estimands are method-specific noise or bias and decision agreement for theoretically relevant contrasts. Participant response style and method-by-item instability are admissible only when the dataset contains the participant-level rows and repeated structure needed to estimate them. This ordering keeps the analysis from selecting whichever scale looks best after the fact, the kind of forking path Gelman and Loken warn about (Gelman & Loken, 2013, 2014).

The substantive comparison is a common-latent-signal null: magnitude estimation, Likert ratings, forced choice, and Thurstonian choice are first modelled as noisy observations of the same item-level acceptability. A magnitude-estimation term becomes interesting only if it improves prediction, reliability, or decision stability enough to matter. Scrambled labels, method-label permutations, and simulations from a no-method-advantage latent model are negative controls for the pipeline, checking whether the analysis manufactures a method winner from degraded structure. They are not substitutes for the substantive common-signal comparison.

4 SCALE AND RELIABILITY

The scale question should be phrased as practical measurement advantage over the common-latent null. Bard et al.'s case for magnitude estimation included scale resolution, fine-grained distinctions, naive-participant use, and replication across groups (Bard et al., 1996). The safer target is that source-

grounded claim, together with the stronger reading of magnitude estimation that took hold in parts of the later methodology literature. Later method-comparison data give the paper a sharper question: does magnitude estimation recover item differences, participant agreement, or theoretically relevant decisions that simpler tasks lose?

Sprouse et al. (2013) compare formal judgment tasks with informal published judgments, giving evidence about convergence between magnitude estimation, Likert ratings, and forced choice. The public Sprouse rows can support a bounded reanalysis of item-level agreement and decision consequences, subject to reuse permissions. Langsford et al. (2018) add a reliability and response-bias target by comparing Likert, magnitude estimation, forced choice, and Thurstonian models. Without their raw rows, Langsford can support the synthesis of published reliability findings but not a new participant-level response-style or test-retest model.

The reliability section should inherit the analysis charter rather than reopen method choices. In the 2013 data, pair-level forced choice is primary and the item-level logistic representation is sensitivity. In the 2017 data, yes/no rows with non-material IDs B, G, and M are outside the item-level comparison set. These choices make later reliability claims traceable to fixed units of analysis instead of to whichever representation later proves most convenient.

The primary reliability analysis should compare methods on prediction, stability, decision consequences, mean separation, and correlation, with priority on consequences for the linguistic contrast. A useful result would show whether magnitude estimation changes the answer for the linguistic contrast under study, reduces Type S or Type M risks (Gelman & Carlin, 2014), or carries costs that offset any added resolution. Sensitivity checks should vary transformations and exclusion rules as named specifications rather than as invisible post hoc choices (Steege et al., 2016).

5 THE CONTEMPORARY AFTERLIFE OF ACCEPTABILITY

TODO: Treat modern benchmark resources as an afterlife of acceptability, not as direct magnitude-estimation evidence. Resources such as BLiMP (Warstadt et al., 2020) and the Syntactic Acceptability Dataset (Juzek, 2024) show how acceptability became portable for computational evaluation, but they also change the object being measured.

6 GRAMMATICALITY AND ACCEPTABILITY

TODO: Connect the reassessment to the broader *(Un)grammatical* program. The paper should separate personal production, acceptability, naturalness, standardness, correction behaviour, and theoretical grammaticality. Existing datasets may show the boundaries of what no-new-data work can settle.

7 CONCLUSION

TODO: Conclude with the balanced update. Bard et al. were right to make acceptability measurable and methodologically serious. The case for magnitude estimation as a solution to scale-resolution problems is a separate, source-grounded claim. The later, stronger reading of magnitude estimation as the privileged measurement tool should be reassessed as reception unless Bard is cited directly for it.

DATA AND CODE AVAILABILITY

No raw participant-level data are committed to this repository. Source and dataset verification are tracked in `notes/source-verification.md`; the analysis charter and structural Sprouse inventory are tracked in `notes/analysis-charter.md` and `notes/dataset-inventory.md`. Scripts currently generate ignored local structural derivatives and readiness manifests, while novel outcome analysis remains gated pending Sprouse data reuse permission or an explicitly verification-only framing.

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TODO: Add final AI-use disclosure with the actual model list used on the project.

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